ROCKET for Univariate Time Series Classification: Experimental Evaluation Against State-of-the-Art Baselines

Alessandro, Anton, Lara, Iñigo
Faculty of Informatics / UPV
San Sebastián, Spain

***Abstract*—In this paper, we present ROCKET, a univariate time series classification algorithm based on random convolutional kernels, which combines speed and high accuracy. We evaluate ROCKET on a selection of datasets from the UCR repository and compare its performance against seven benchmark algorithms: 1NN-Euclidean, 1NN-DTW, Shapelet Transform, BOSS ensemble, Time Series Forest, InceptionTime and catch22. Our experimentation employs a robust validation framework and considers multiple evaluation metrics, including accuracy and computational efficiency. The results demonstrate that ROCKET consistently outperforms the benchmark algorithms in both accuracy and scalability, demonstrating its effectiveness for large-scale time series classification tasks.**

***Index Terms*—Time series classification, Random convolutional kernels**

I. Introduction

Time series classification is a fundamental task in many domains where large volumes of sequential data are generated and need to be processed efficiently and accurately. Examples include financial markets, healthcare and industrial monitoring. Traditional approaches often achieve competitive accuracy but suffer from high computational cost and limited scalability when applied to large or complex datasets [1], [2].

ROCKET (RandOm Convolutional KErnel Transform) is an algorithm designed to overcome these limitations. It applies a large number of randomly initialized convolutional kernels to transform time series data into feature representations, which are then fed into linear classifiers. This approach enables both fast processing and high classification accuracy, even on large-scale datasets.

In this paper, we evaluate ROCKET on a selection of UCR datasets [3] and compare its performance with six benchmark algorithms, analyzing both accuracy and computational efficiency.

II. ROCKET Method

ROCKET transforms time series by applying convolutional kernels, which are not learned through training, but instead are randomly initialized [4]. Each kernel has parameters including length, weights, dilation, and bias, all of which are assigned randomly. The randomness allows the algorithm to generate a diverse set of filters, capable of capturing a wide variety of temporal patterns.

Once the kernels are applied to a time series, each convolution produces a feature map, showing the activation of the kernel across the entire series. From each feature map, ROCKET extracts two key summary features: Maximum value and Proportion of Positive Values (PPV).

Using a large number of kernels, ROCKET transforms each time series into a high-dimensional feature vector, which captures a large variety of temporal patterns without requiring kernel training.

These feature vectors are then fed into a linear classifier, typically Ridge Regression or Logistic Regression, which performs the final classification.

A. Kernels

ROCKET relies on a large number of convolutional kernels to extract meaningful features from time series. Each kernel acts like a small filter that highlights certain patterns in the data. Although the kernels are randomly initialized and not learned, their diversity allows the algorithm to extract rich and varied features from the time series. This random initialization enables ROCKET to efficiently explore a wide variety of temporal patterns without the computational cost of training each kernel.

The main parameters of each kernel are:

- Length l: determines how many consecutive points the kernel considers, which allows kernels of different scales to detect both short and long-term patterns.
- Weights w: sampled from a standard normal distribution, giving each kernel a unique pattern of emphasis over the input series.
- Dilation d: controls the spacing between points in the kernel. A dilation of $d > 1$ allows the kernel to skip points in the series, allowing the model to capture patterns that occur over longer time intervals or at multiple scales.
- Bias b: random offset added to the convolution output, which increases variability of the resulting feature map.

Additionally, padding is applied when necessary to ensure that kernels cover the full series, preserving feature extraction across all time points. Each kernel produces a feature map along the series, which will later be summarized in the feature extraction stage.

B. Time Series Transformation

C. Feature Extraction

D. Classifier

E. Advantages

F. Limitations

III. Variants

Since its introduction, ROCKET has inspired a family of successors. These methods address its main limitations: computational overhead, non-determinism, and restriction to univariate input. At the same time, they preserve or improve classification accuracy. The three principal variants are described below.

A. MiniROCKET

MiniROCKET (Dempster et al., 2021) is designed to be almost deterministic and substantially faster than the original, while achieving accuracy that is statistically indistinguishable from ROCKET across the UCR archive.

The key architectural change is the restriction of kernel weights to elements of the set $\{-1, +1\}$, drawn from a fixed, small pool of weight vectors rather than sampled afresh from a standard normal distribution. Specifically, MiniROCKET enumerates all possible binary weight vectors of lengths $(7, 9, 11)$ whose entries belong to $\{-1, +1\}$ and whose number of positive entries satisfies a fixed proportion, then selects 84 of them. These 84 weight vectors are reused across all kernels in the bank; only dilation and bias vary. Crucially, bias values are no longer random: they are drawn systematically from the quantiles of the convolution outputs observed on the training data. This data-dependent bias calibration is the sole non-deterministic step in MiniROCKET, and its effect on run-to-run variance is negligible in practice.

The impact on speed is significant. Because the weights are binary and fixed, the sliding dot-product computation can be replaced by simple integer additions and subtractions, reducing the constant factor of the transform by approximately $4\times$ compared to ROCKET. On the full 85-dataset UCR benchmark, MiniROCKET trains in roughly one quarter of the time required by ROCKET while achieving a mean accuracy within 0.01 percentage points of ROCKET's mean accuracy. MiniROCKET retains both output features of the original — PPV and max — though empirical results show that PPV alone is nearly sufficient, consistent with the sensitivity analysis reported in the original ROCKET paper.

B. MultiROCKET

MultiROCKET (Tan et al., 2022) is the principal extension of the ROCKET family to multivariate time series and to a richer feature vocabulary.

On the feature-extraction side, MultiROCKET augments the two original summary statistics — PPV and max — with three additional pooling operators applied to each feature map:

- **Mean of Positive Values (MPV)**: the average activation strength at positions where the convolution output is positive, capturing how strongly a pattern matches when it does match.
- **Mean of Indices of Positive Values (MIPV)**: the average position in the series at which the kernel produces a positive output, encoding where in time the matched pattern tends to occur.
- **Longest Stretch of Positive Values (LSPV)**: the length of the longest contiguous run of positive outputs, providing a measure of the temporal persistence of a given pattern.

These three operators together with PPV and max yield $5 \times k$ features per channel per series, allowing the linear classifier to exploit not only whether and how strongly a pattern appears, but also where and for how long it persists. Across the UCR archive, the inclusion of these additional features yields a statistically significant improvement in mean rank over ROCKET.

For multivariate series with c channels, MultiROCKET applies the transform independently to each channel and concatenates the resulting feature vectors, producing $5 \times k \times c$ features in total. While this design ignores cross-channel interactions, it preserves the linear-time complexity of the original transform and avoids the combinatorial explosion that would result from applying kernels to all pairs of channels. MultiROCKET also inherits MiniROCKET's weight scheme, using the same fixed pool of 84 binary weight vectors, so the overall computational complexity is $O(k \cdot n \cdot l \cdot c)$, linear in all four quantities.

C. Hydra

Hydra (Dempster et al., 2023) represents a conceptually distinct evolution of the ROCKET paradigm, combining ideas from random convolutional kernels with dictionary-based methods.

Rather than summarising each feature map with scalar pooling statistics, Hydra organises kernels into groups of g kernels sharing the same dilation and padding, and builds a histogram of nearest-kernel assignments across each group and each position in the series. This histogram — counting, for each kernel within a group, how many time-series positions are most similar to that kernel's weights — serves as the feature representation passed to the linear classifier.

This construction is closely related to the Bag-of-SFA-Symbols (BOSS) dictionary representation: the group of g kernels can be seen as a soft codebook, and the histogram as a symbolic frequency profile. The key difference is that Hydra's codebook is random and extremely cheap to compute, whereas BOSS requires a Symbolic Fourier Approximation discretisation step that is quadratic in training-set size. Empirically, Hydra surpasses ROCKET and MultiROCKET in mean classification rank on the UCR archive while remaining an order of magnitude faster than InceptionTime. It is particularly effective on problems with long and complex series, where the positional frequency of matched patterns — rather than just their peak value or overall proportion — carries the most discriminative information.

A combined variant, **Hydra-MultiROCKET**, concatenates Hydra's histogram features with MultiROCKET's pooling features, achieving the highest mean accuracy in the ROCKET family to date on the UCR archive.

IV. Experimentation

V. Results and Discussion

VI. Conclusions

References

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